Elementary Analysis

Section 32: The Riemann Integral

Exercises 32.1–32.8, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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32 The Riemann integral

For a bounded function f and a partition $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$, let $I_j = [t_{j-1}, t_j]$. The lower and upper sums are

$$L(f,P) = \sum_j (\inf_{I_j} f)\,|I_j|\,, \qquad U(f,P) = \sum_j (\sup_{I_j} f)\,|I_j|\,,$$

where $|I_j| = t_j - t_{j-1}$ denotes interval length. The lower integral is the supremum of all lower sums; the upper integral is the infimum of all upper sums. Write these as $\underline{I}(f)$ and $\overline{I}(f)$. Integrability means they are equal. A practical equivalent test is that $U(f,P) - L(f,P)$ can be made arbitrarily small. No antiderivative formula is assumed in the Darboux calculations below.

32.1 Exercise 32.1: The integral of a cube from sums

What we must show. Compute both Darboux integrals of x^3 on $[0,b]$, where $b > 0$.

Choose equal subintervals and identify maxima and minima.

The cube is increasing on $[0,b]$. For $t_k = kb/n$, its maximum on $[t_{k-1}, t_k]$ is t_k^3 and its minimum is t_{k-1}^3. Therefore

$$U_n = \frac{b^4}{n^4}\sum_{k=1}^{n} k^3 = \frac{b^4(n+1)^2}{4n^2}, \qquad L_n = \frac{b^4}{n^4}\sum_{k=0}^{n-1} k^3 = \frac{b^4(n-1)^2}{4n^2}.$$

Here we used $\sum_{k=1}^{n} k^3 = [n(n+1)/2]^2$.

Squeeze the integrals.

For every n, $L_n \le \underline{I}(f) \le \overline{I}(f) \le U_n$. Both outer sequences tend to $b^4/4$, because $(n \pm 1)^2/n^2 \to 1$. Thus the two integrals coincide at that value.

Conclusion. The function is integrable and $\int_0^b x^3\,dx = b^4/4$.

32.2 Exercise 32.2: A rational-only linear function

What we must show. For $f(x) = x$ on rational points and $f(x) = 0$ on irrational points of $[0,b]$, compute its lower and upper integrals.

Use density on each interval.

Every nondegenerate interval $[u,v] \subseteq [0,b]$ contains irrational points, so its infimum is zero. Its rational points can approach v from below; their values approach v, while no value exceeds v. Thus its supremum is v, whether or not v itself is rational. For any partition, consequently,

$$L(f,P) = 0, \qquad U(f,P) = \sum_{j=1}^{n} t_j(t_j - t_{j-1}).$$

In particular, the lower integral is 0.

Find the infimum of the upper sums.

Since $t_j \geq t_{j-1} \geq 0$,

$$t_j(t_j - t_{j-1}) \geq \frac{t_j + t_{j-1}}{2}(t_j - t_{j-1}) = \frac{t_j^2 - t_{j-1}^2}{2}.$$

Summing telescopes to $U(f, P) \geq b^2/2$. For the equal partition,

$$U(f, P_n) = \frac{b^2}{n^2} \sum_{j=1}^{n} j = \frac{b^2(n+1)}{2n} \longrightarrow \frac{b^2}{2}.$$

An infimum bounded below by $b^2/2$ and approached by such sums equals $b^2/2$.

Conclusion. (a) The lower integral is 0 and the upper integral is $b^2/2$. (b) They differ for $b > 0$, so f is not integrable.

32.3 Exercise 32.3: A rational-only quadratic function

What we must show. Repeat the calculation for $g(x) = x^2$ on rational points and $g(x) = 0$ on irrational points of $[0, b]$.

Identify each interval's extrema.

Density and nonnegativity give $\inf_{[u,v]} g = 0$ and $\sup_{[u,v]} g = v^2$. Thus all lower sums vanish and $U(g, P) = \sum_j t_j^2(t_j - t_{j-1})$.

Compute the infimum of these upper sums.

For $0 \leq u \leq v$ we have $v^2 \geq (v^2 + vu + u^2)/3$, whence

$$v^2(v - u) \geq \frac{v^3 - u^3}{3}.$$

This telescopes to $U(g, P) \geq b^3/3$ for every P. For equal subintervals,

$$U(g, P_n) = \frac{b^3}{n^3} \sum_{j=1}^{n} j^2 = \frac{b^3 n(n+1)(2n+1)}{6n^3} \longrightarrow \frac{b^3}{3}.$$

Conclusion. (a) The lower and upper integrals are 0 and $b^3/3$. (b) The function is not integrable on $[0, b]$ for $b > 0$.

32.4 Exercise 32.4: Refinement by induction

What we must show. Extend the one-point refinement argument in Lemma 32.2 to any finite refinement $P \subseteq Q$.

State precisely what is being inducted on.

Let r be the number of points of Q not already in P. We claim

$$L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P).$$

For $r = 0$ the partitions are equal; the middle inequality follows because each interval's infimum is at most its supremum.

Insert one more point.

Suppose the result is established for r extra points. If Q has $r+1$ extra points, choose one of them, q, and set $Q' = Q \setminus \{q\}$. By the induction hypothesis, $L(f,P) \le L(f,Q')$ and $U(f,Q') \le U(f,P)$. The proved one-point case, applied to Q' and Q, gives $L(f,Q') \le L(f,Q)$ and $U(f,Q) \le U(f,Q')$. Combining the inequalities proves the two outer comparisons for Q. The middle comparison always holds interval by interval.

Conclusion. Adding finitely many partition points can only increase a lower sum and decrease an upper sum.

32.5 Exercise 32.5: Why the lower integral cannot exceed the upper

What we must show. Apply the order property of suprema and infima to Theorem 32.4.

Name the two sets.

Let

$$S = \{L(f,P) : P \text{ is a partition of } [a,b]\}, \qquad T = \{U(f,Q) : Q \text{ is a partition of } [a,b]\}.$$

Both are nonempty and bounded because f is bounded and the total interval length is $b-a$.

Compare sums from different partitions.

For any P, Q, their union $R = P \cup Q$ refines both. Lemma 32.2 gives

$$L(f,P) \le L(f,R) \le U(f,R) \le U(f,Q).$$

Thus every $s \in S$ is at most every $t \in T$. By Exercise 4.8, $\sup S \le \inf T$. To see the order reasoning directly, each $t \in T$ is an upper bound for S, so $\sup S \le t$ for all $t \in T$; therefore $\sup S$ is a lower bound for T, and hence is at most $\inf T$.

Conclusion. $\underline{I}(f) = \sup S \le \inf T = \overline{I}(f)$.

32.6 Exercise 32.6: Upper and lower sums whose gap tends to zero

What we must show. Prove integrability and identify the limits of the given sums. The sums U_n and L_n need not come from the same partition. Nevertheless, the cross-partition comparison just established gives

$$L_n \le \underline{I}(f) \le \overline{I}(f) \le U_n.$$

Subtracting shows $0 \le \overline{I}(f) - \underline{I}(f) \le U_n - L_n \to 0$. The fixed middle difference must therefore be zero. Write the common integral as I. Then

$$0 \le U_n - I \le U_n - L_n, \qquad 0 \le I - L_n \le U_n - L_n.$$

Each right-hand side tends to zero, so both $U_n \to I$ and $L_n \to I$.

Conclusion. f is integrable and $\int_a^b f = \lim_n U_n = \lim_n L_n$.

32.7 Exercise 32.7: Changing finitely many values

What we must show. Show that changing an integrable function at finitely many points preserves its integral.

First understand the finite-support error.

Put $h = g - f$. It is zero except at finitely many points $c_1, \dots, c_m$, so it is bounded; choose $H > 0$ with $|h| \le H$. Given any $\eta > 0$, make a partition that contains the exceptional points and nearby points on each available side, so the total length of cells meeting these points is less than η. This is possible because there are only finitely many points. If a point is a or b, only its one adjacent side is needed. On every other cell $h = 0$; on the exceptional cells $-H \le h \le H$. Thus

$$-H\eta \le L(h, P) \le U(h, P) \le H\eta.$$

This proves that a finite-support function has lower and upper integrals both zero, by letting $\eta \downarrow 0$.

Combine the estimate with the original integrable function.

Here is a direct argument that does not assume a later linearity theorem. Given $\varepsilon > 0$, choose a partition for f with upper-minus-lower gap less than $\varepsilon/2$. Refine it to the partition just described, with exceptional-cell total length less than η and $2H\eta < \varepsilon/2$. Refinement preserves the bound for f. On any cell J, $g = f + h$ implies

$$\sup_J g \le \sup_J f + \sup_J h, \qquad \inf_J g \ge \inf_J f + \inf_J h.$$

Consequently $U(g, P) - L(g, P) < \varepsilon/2 + 2H\eta < \varepsilon$. This proves g is integrable.

Compare the two integral values.

For the same partition, $L(g, P) \ge L(f, P) - H\eta$ and $U(g, P) \le U(f, P) + H\eta$. Both integrals lie between the respective sums. Hence their difference in absolute value is at most $U(f, P) - L(f, P) + H\eta$. The original gap and η can both be made arbitrarily small.

Conclusion. $\int_a^b g = \int_a^b f$. Isolated values do not change area.

32.8 Exercise 32.8: Restriction to a smaller interval

What we must show. Prove integrability of f on $[c, d] \subseteq [a, b]$. Assume $c < d$; the zero-length case has integral zero by convention. Given $\varepsilon > 0$, choose a partition P of $[a, b]$ with $U(f, P) - L(f, P) < \varepsilon$. Add c and d to form the refinement Q. Its gap is still less than ε. The points of Q lying in $[c, d]$ form a partition Q_0 of that interval. Its gap is a sub-sum of the gap for Q:

$$U_{[c,d]}(f, Q_0) - L_{[c,d]}(f, Q_0) = \sum_{J \subseteq [c,d]} (\sup_J f - \inf_J f)\,|J| \le U_{[a,b]}(f, Q) - L_{[a,b]}(f, Q).$$

Every term omitted on the right is nonnegative. Thus the smaller interval has partitions with gap below every positive ε.

Conclusion. The Darboux criterion proves that $f|_{[c,d]}$ is integrable.